AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

INDIVIDUAL THESIS

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ABSTRACT

This thesis examines a landmark recognition and intelligent assistant component for a unified Sri Lankan tourism platform. A recorded notebook compares a custom convolutional network, fine-tuned MobileNetV2, a native Vision Transformer, and a support-vector classifier using deep features. The loader reports 5,500 images across 17 classes, partitioned into 4,400 training and 1,100 validation images. The deep-feature SVM achieves the highest recorded validation accuracy of 93.18 percent and weighted F1 of 93.23 percent. The validation partition is reused during model development, so these are not independent final-test estimates. The application links predicted labels to structured landmark metadata and contextual conversation. Gemini provides generated responses when available, while local rules answer common visitor questions during provider failure. Inspection identifies differences between training and serving feature extraction and preprocessing, as well as inconsistent accuracy labels. These prevent the notebook metric from being treated as verified deployed performance. The individual contribution is a connected recognition, information-retrieval, and assistant workflow with a comparative recognition experiment. Independent image testing, data-provenance verification, serving-parity checks, and assistant factuality evaluation remain necessary.

Keywords: landmark recognition, MobileNetV2, support-vector machine, tourism, contextual assistant

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
CNN	Convolutional neural network
CSV	Comma-separated values
LLM	Large language model
RGB	Red green blue
SVM	Support-vector machine
TFLite	TensorFlow Lite
ViT	Vision Transformer

1 INTRODUCTION

1.1 Overall project context

The group research project, AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration, brings several tourism functions into one application. This individual component addresses a visitor who can photograph a place but may not know its name or cultural context. The workflow links image recognition to landmark information and a contextual assistant. Route and vehicle advice, itinerary generation, and accommodation functions remain adjacent team contributions.

1.2 Individual component context

AI Landmark Recognition & Tourist Information combines two connected tasks: identifying a supported landmark from an image and answering questions about that landmark. The implementation includes an image classifier, metadata lookup, visitor-information display, and the Ayubowan AI chat interface. The report examines the training notebook, deployed inference code, and frontend integration as separate sources of evidence [1], [2], [3].

1.3 Problem statement

A photograph does not directly provide a searchable place name. Even after identification, a visitor may need historical background, visiting hours, admission information, etiquette, and nearby facilities. Requiring separate image search and repeated text queries interrupts this process. The technical problem is to connect visual identity to useful, context-specific information while handling uncertain predictions and unavailable language-model services. An incorrect identity can also misdirect every subsequent answer, so recognition and assistance must be assessed together.

1.4 Research gap

The practical gap addressed here is continuity between recognition and conversation inside the group application. The assistant should inherit the recognised landmark instead of asking the visitor to re-enter it. This is an integration contribution, not a

claim that image recognition or conversational tourism systems are new. A broader systematic comparison of existing products and user needs has not been established by the available project evidence.

1.5 Aim and objectives

The aim is to develop and evaluate a landmark recognition component with contextual visitor assistance for supported Sri Lankan locations.

The objectives are to prepare labelled images; compare a custom CNN, MobileNetV2 transfer learning, a native Vision Transformer, and a deep-feature SVM; export inference artefacts; associate predictions with structured landmark information; provide contextual chat with a local fallback; and identify limitations in evaluation and deployment consistency.

1.6 Research questions or hypotheses

RQ1: Which of the four implemented architectures achieves the strongest recorded classification performance on the notebook validation partition?

RQ2: How can the recognised class be connected to relevant visitor information and conversational assistance?

RQ3: Which differences between training and serving limit the transfer of notebook results to the running application? These questions concern implemented behaviour and recorded experiments rather than a tested hypothesis about tourist satisfaction.

1.7 Scope limitations and assumptions

Recognition is restricted to 17 labels. The metadata CSV contains 381 rows, but this does not mean that all 381 places are supported by the image classifier. The component supports still-image upload and landmark-related conversation. It does not implement continuous video recognition, visual localisation, unrestricted recognition of all Sri Lankan attractions, or navigation.

The implementation assumes that class labels and metadata names can be mapped consistently. Source-image permissions, duplicate control, and metadata currency require further documentation. No field study or independent tourist evaluation is claimed.

1.8 Individual contribution

The individual scope of Madushan M. (IT22214720) is the landmark recognition and intelligent assistant component: training and comparing recognition models, connecting deployed inference to landmark metadata, and presenting recognition and conversational outputs. Authorship of individual commits and shared integration responsibilities should be checked against the agreed group allocation. The report does not attribute the route, vehicle, itinerary, or accommodation modules to this member.

1.9 Thesis structure

Chapter 2 reviews methodological foundations. Chapter 3 describes the dataset and experiment. Chapters 4 and 5 explain design and implementation. Chapter 6 presents stored results and notebook figures. Chapters 7 and 8 discuss interpretation, deployment gaps, and future work. Appendices document reproducibility and report-preparation assistance.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This focused narrative review uses primary research on mobile convolutional networks, support-vector classification, and vision transformers. The sources explain the implemented model families; the project notebook and source files establish what was actually built. No systematic search protocol, inclusion-count flow, or exhaustive tourism-system review is claimed.

2.2 Domain and theoretical foundations

Image classification maps an image to one of a predefined set of labels. A CNN learns spatial features, whereas transfer learning reuses features obtained from a larger source task. MobileNetV2 uses inverted residual blocks and linear bottlenecks to support efficient visual representation [4]. In this project, its convolutional representation is used both with a neural classification head and with a separate SVM.

A support-vector classifier learns a decision boundary in feature space [5]. The project applies an RBF kernel to scaled deep features. Its probability output is a model score used by the interface, not a guarantee that a recognised place is correct.

2.3 Related research

Sandler et al. motivate a compact convolutional architecture suitable for constrained inference [4]. Cortes and Vapnik provide the foundation for support-vector networks [5]. Dosovitskiy et al. demonstrate image classification using sequences of image patches and large-scale pretraining [6]. These works support the choice of contrasting architectures.

The project's native ViT is trained under its own small-data configuration. Its result cannot be interpreted as a general rejection of transformers or as a direct replication of a large pretrained ViT. Likewise, the CNN and transfer-learning models differ in prior training and capacity, so the comparison is between the actual configurations rather than isolated architecture effects.

2.4 Existing systems products and techniques

The local alternatives are a custom CNN, a fine-tuned MobileNetV2, a native ViT, and a deep-feature SVM. They share the notebook validation partition but differ in feature learning and classifier design. The assistant uses deterministic CSV retrieval followed by Gemini text generation when available. A keyword-driven response function provides an alternative when the service is unavailable.

Although GraphRAG appears in the group research title, the inspected landmark endpoint does not traverse a graph or retrieve vector embeddings. Its evidenced retrieval mechanism is class-name mapping and literal CSV name matching [2].

2.5 Gap synthesis and conceptual framework

The conceptual workflow is image acquisition, classification, metadata retrieval, visitor-information display, and contextual conversation. Recognition establishes the proposed identity; the metadata supplies factual context; the assistant translates a question and context into a response. Each boundary needs separate evaluation. Classification accuracy alone cannot establish answer correctness, factual currency, or successful visitor decision-making.

3 RESEARCH METHODOLOGY

3.1 Research design

The research follows a prototype-development approach with a comparative image-classification experiment. The notebook records data loading, model training, validation, plots, and export [1]. Implementation inspection establishes the API and frontend design [2], [3]. Stored outputs are reported as historical experiment evidence; no model retraining or new live-service benchmark was performed during report preparation.

3.2 Data participants or study materials

The notebook loader reports 5,500 images, with 4,400 training and 1,100 validation images across 17 classes. A separate directory-count table totals 5,501 files, producing a one-file discrepancy. The loader counts are used for the experiment description, while the difference must be reconciled against a frozen dataset manifest.

The class-count figure shows imbalance, from 159 Lankatilaka Vihara files to 548 Sigiriya files in that directory listing. These are notebook inventory counts rather than a verified current dataset census. The image source collection, licences, capture dates, and duplicate-removal process are not documented sufficiently to establish provenance.

Landmark descriptions are held separately in Sri_Lankan_Landmark.csv, with 381 rows and fields for name, location, history, admission, hours, and nearby facilities [7].

3.3 Experimental or study procedure

The notebook uses seed 42, 224 by 224 images, batch size 32, and an 80/20 directory split. The training pipeline applies horizontal flips, rotation, zoom, translation, and contrast augmentation. Augmentation occurs before caching, so the first cached augmented representation may be reused rather than regenerated every epoch. Balanced class weights are computed for neural-network training.

The custom CNN uses convolutional blocks, batch normalisation, pooling, dropout, and a dense classification head. MobileNetV2 starts from ImageNet weights with a frozen backbone; fine-tuning subsequently enables layers from index 100 with a lower learning rate. Both stages allow up to 15 epochs and use validation-based callbacks.

The ViT divides images into 16-pixel patches and uses a 128-dimensional projection, four attention heads, and four transformer blocks. The SVM receives 1,280-dimensional features from the fine-tuned MobileNetV2 global-average-pooling layer. StandardScaler is fitted on training features; the classifier uses an RBF kernel, C of 1.0, probability estimation, and seed 42.

3.4 Measures and evaluation metrics

Accuracy is the proportion of correctly predicted validation labels. Precision, recall, and F1 are reported using support-weighted aggregation in the comparison table. The notebook also provides per-class reports and a row-normalised confusion matrix. Weighted summaries should be read alongside class-level errors because common classes contribute more strongly.

The validation set is used during neural training for early stopping and checkpoint selection and later for model comparison. It is therefore not an untouched final test set. Reported percentages describe this reused partition and may overestimate independent generalisation.

3.5 Data analysis

The comparison uses the stored summary table and plots without reconstructing unrecorded trials. The highest recorded accuracy is selected descriptively; no repeated-seed uncertainty interval or significance test is available. Latency values come from notebook loops around prediction calls and include framework overhead. They are not mobile-device, network, or chatbot response-time measurements.

The notebook SVM file size excludes the feature extractor and scaler. Its training duration excludes prior backbone training and feature extraction. The MobileNet

training timer is attached to the first training stage, so its recorded duration does not represent the complete two-stage procedure.

3.6 Ethics privacy security and responsible AI

Image collection and metadata reuse require documented permissions and source attribution. Uploaded photographs may include people or location clues, so production deployment should define retention, access, and deletion behaviour. The recognition function processes image bytes in memory; that observation alone does not establish a complete privacy assessment.

The chat function can transmit the selected metadata, user message, and recent history to an external provider. Users should be informed of that boundary. Cultural and practical visitor information should be reviewed by qualified sources, and time-sensitive entries should include dates.

3.7 Reproducibility and research artefacts

Reproducibility depends on the exact dataset manifest, split membership, notebook environment, saved feature extractor, scaler, classifier, label order, and exported model. The notebook records TensorFlow 2.20.0 and GPU availability. The raw DataNew image collection is not available in the inspected project files, so this report does not reproduce training.

The provided figures are extracted directly from saved notebook outputs. They are existing experiment artefacts, not screenshots of the current application and not newly generated results.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component must accept an image, return a supported landmark identity or an uncertain result, retrieve matching metadata, and open a contextual assistant. The assistant must accept a message, retain limited history, and return a response when the external provider is unavailable. Non-functional needs include bounded requests, consistent preprocessing, readable information, and clear uncertainty.

Condition	Implemented response
Missing image or empty filename	HTTP 400 with an explanatory error
Declared request size above 10 MB	HTTP 413
Invalid image bytes	HTTP 400
Confidence below 40 percent	Success response marked unrecognized
SVM unavailable after loading attempt	TFLite inference path
Missing Gemini key or request failure	Local rule-based response

Table 4.1. Source-inspected handling of recognition and assistant conditions.

4.2 Component architecture within the complete solution

The React Native/Expo interface sends multipart image data to the Flask AI service. The default SVM path extracts MobileNetV2 features, scales them, and obtains class probabilities. The alternate path invokes the quantized TFLite classifier under a lock. A predicted label is mapped to a CSV name, and the first matching row becomes structured metadata.

Chat is a separate JSON endpoint. It receives a message, landmark identity, and history, retrieves metadata, and calls the response function. The frontend passes recognised class context into the chat screen. The architecture keeps image inference and language generation distinct while linking them through the landmark identity.

4.3 Design alternatives and decisions

The SVM path offers the strongest recorded notebook validation result but requires both a feature extractor and a classifier. TFLite offers a smaller exported neural

model and a fallback deployment path. A shared label file keeps class indices interpretable.

CSV lookup is simple and inspectable, but first-match substring retrieval can be ambiguous and does not provide ranked evidence. A rule-based assistant keeps common factual queries available without Gemini, but it handles a restricted set of keyword categories and lacks full conversational reasoning.

4.4 Detailed design

Images are decoded with Pillow, converted to RGB, resized to 224 by 224, and converted to a float32 batch. The prediction endpoint chooses an engine, obtains the highest-scoring label, and rounds confidence to two decimal places. Scores below 40 percent return an unrecognised result rather than metadata. This threshold is a heuristic; an open-set rejection study is not recorded.

Metadata lookup uses a fixed mapping from 17 model labels to CSV search strings. It selects the first literal case-insensitive name match. If chat lacks a recognised context, it searches the user message for supported names. This is deterministic name retrieval, not a learned retriever.

The external chat request includes non-empty metadata fields and up to six previous messages. Its timeout is eight seconds. The local fallback checks price, time, nearby facilities, history, and etiquette keywords, then returns an overview when no category matches. The frontend uses a separate 30-second request timeout.

4.5 Standards and professional practice

The frontend and backend exchange JSON and multipart data through named endpoint contracts. Errors should remain understandable without exposing internal exception details. In the current code, some generic failures return exception strings; production hardening would require controlled error messages.

The phrase VERIFIED LANDMARK KNOWLEDGE BASE appears in the prompt, but a prompt label does not independently verify CSV facts. The report treats those fields as project metadata. Secure credential handling, metadata governance, and accessibility should be tested before deployment beyond the prototype.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The training environment uses Python, TensorFlow/Keras, NumPy, pandas, scikit-learn, matplotlib, seaborn, and joblib. The service uses Flask, Pillow, requests, TensorFlow Lite, and saved SVM artefacts. The frontend uses React Native and Expo. The inference service holds models in process, so startup availability and model compatibility affect all recognition endpoints.

5.2 Component implementation

The source defines predict, chat, list, and health endpoints under /api/landmark/. The health endpoint exposes the class count and SVM availability. The list endpoint returns metadata for supported model labels. The recognition screen supports image selection, an analysis state, an uncertain-result state, information cards, and a Chat Guide action.

The chat screen provides quick prompts for etiquette, foreign admission prices, visiting times, history, restaurants, and hotels. Its initial message adapts to the selected landmark. The displayed engine names include fixed historical accuracy percentages; those strings are not computed evaluation results for the current request.

5.3 Integration with the other components

The component shares application navigation and AI-service connectivity with the wider tourism platform. A selected landmark is passed from the recognition screen into the assistant. This creates continuity between visual discovery and question answering without making the landmark module responsible for itinerary optimisation or route safety.

The project title refers to GraphRAG and LLM orchestration across the platform. Within this component, the evidence supports CSV-grounded prompting and provider fallback. The report does not substitute a graph-based architecture for the inspected code.

5.4 Verification and software testing

Source inspection confirms implemented checks and response branches, but a passing landmark-specific automated test run was not available for this report. The notebook evaluates recognition on its validation partition. Neither source inspection nor those metrics verifies the complete deployed image-to-chat path.

Required integration checks include a known-class image, an unsupported image, corrupt bytes, an oversized request, an unavailable SVM, an unavailable Gemini service, an ambiguous metadata match, and repeated chat turns. Their results should be recorded with model hashes and request identifiers rather than presented as completed tests here.

5.5 Deployment and operation

The service expects label, scaler, SVM, and TFLite files under its models directory and landmark metadata under its data directory. SVM enablement is configurable through LANDMARK_ENABLE_SVM; Gemini credentials and model selection use environment configuration. The backend recreates an ImageNet MobileNetV2 extractor and may require cached or downloadable weights.

The TFLite interpreter is loaded before the optional SVM block. Therefore the fallback model is also a startup dependency: a missing or incompatible TFLite file can prevent module loading even if SVM artefacts exist. Server-side TFLite inference is implemented; on-device mobile inference has not been demonstrated.

6 RESULTS AND VALIDATION

6.1 Research results

The notebook comparison table records the following values on 1,100 validation images [1]. These are stored outputs from one experiment, not independently rerun test results.

Model	Accuracy	Weighted precision	Weighted F1
Custom CNN	49.45%	57.86%	47.57%
Fine-tuned MobileNetV2	85.55%	85.91%	85.54%
Native ViT	42.55%	48.47%	41.55%
Deep features with SVM	93.18%	93.48%	93.23%

Table 6.1. Recorded notebook validation performance.

SVM has the highest recorded accuracy, 7.63 percentage points above MobileNetV2. This is a descriptive difference on the reused validation partition. It does not establish independent test superiority or statistical significance.

Model	Latency in ms	Saved file in MB	Recorded training seconds
Custom CNN	76.13	3.49	477.9
Fine-tuned MobileNetV2	100.63	44.07	175.6
Native ViT	80.12	29.17	135.1
Deep features with SVM	290.91	32.13	74.8

Table 6.2. Notebook timing and file-size observations with unequal measurement scope.

The SVM path is slower in this benchmark despite its higher accuracy. Its file-size and training-time columns exclude substantial dependent work. These figures cannot alone determine total deployment cost.

6.2 Product and technical results

The following figures are original PNG outputs embedded in the training notebook. The inventory plot describes the directory-count stage, the confusion matrix describes SVM validation predictions, and the learning curves describe the neural training stages. The figures document the recognition experiment; implementation behaviour is discussed in Chapter 5.

6.2.1 Dataset inventory

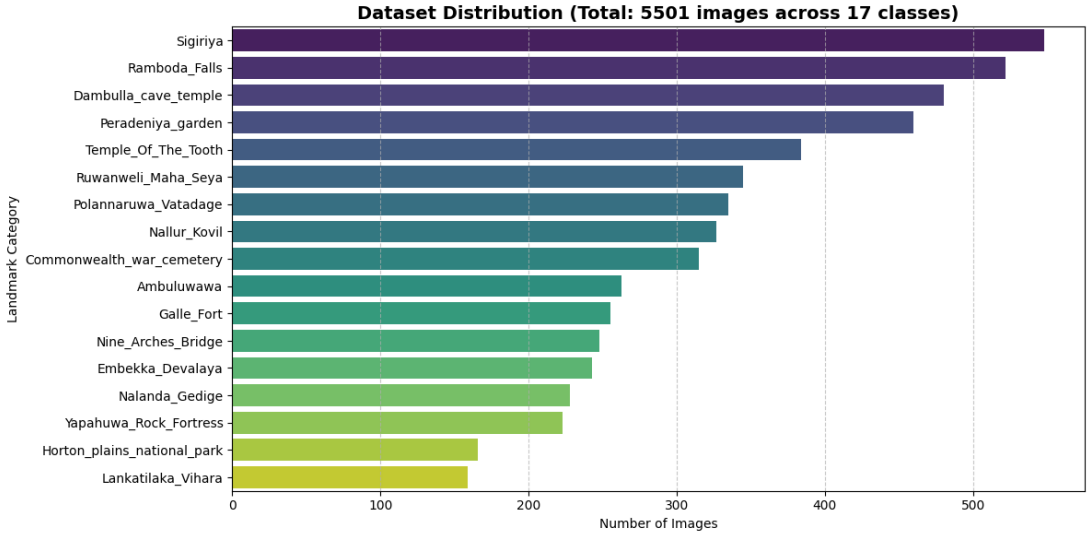


Figure 6.1. Notebook directory-count distribution across the 17 classes.

The inventory chart reflects 5,501 counted files. The loader separately reports 5,500 images; the difference requires a manifest audit. Source: saved notebook output [1].

6.2.2 Class-level recognition

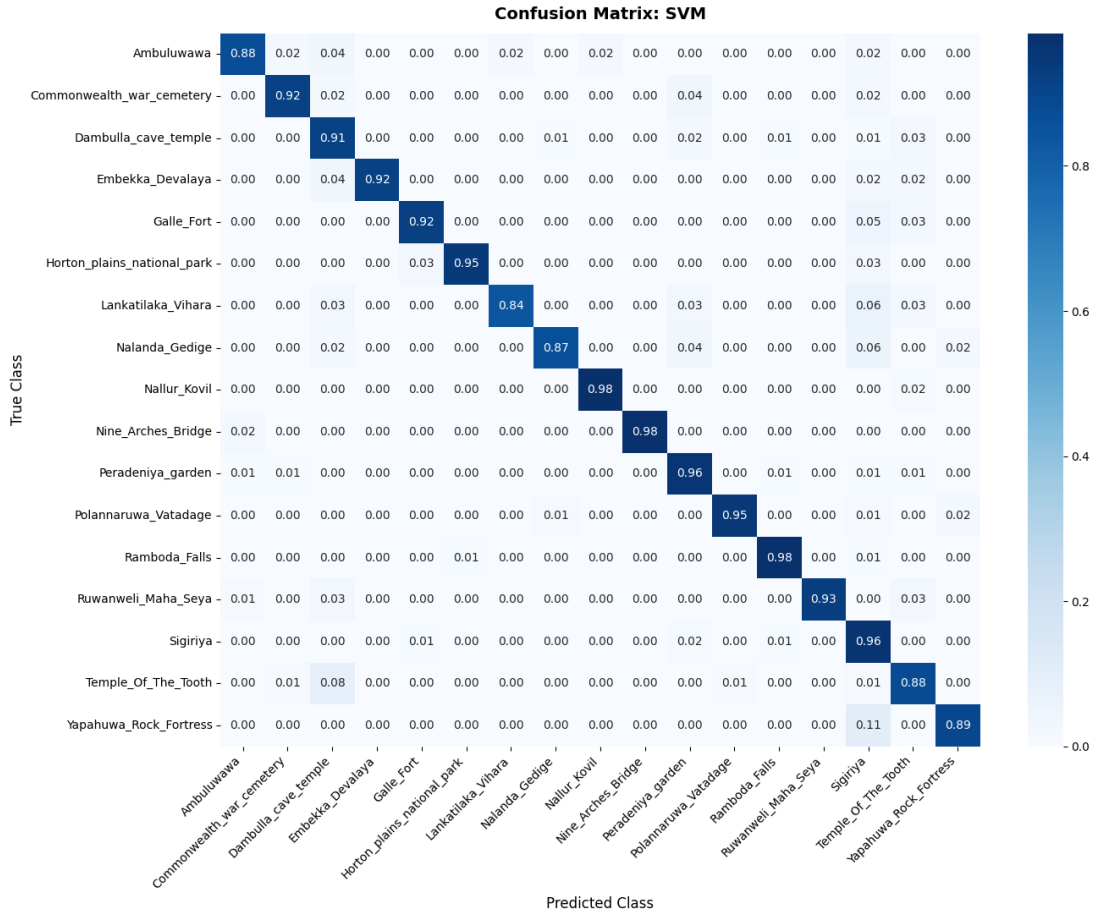


Figure 6.2. Row-normalised SVM confusion matrix from the recorded validation run.

The matrix shows validation confusion among supported labels. It contains no unknown-landmark class and cannot establish rejection quality. Source: saved notebook output [1].

6.2.3 Training behaviour

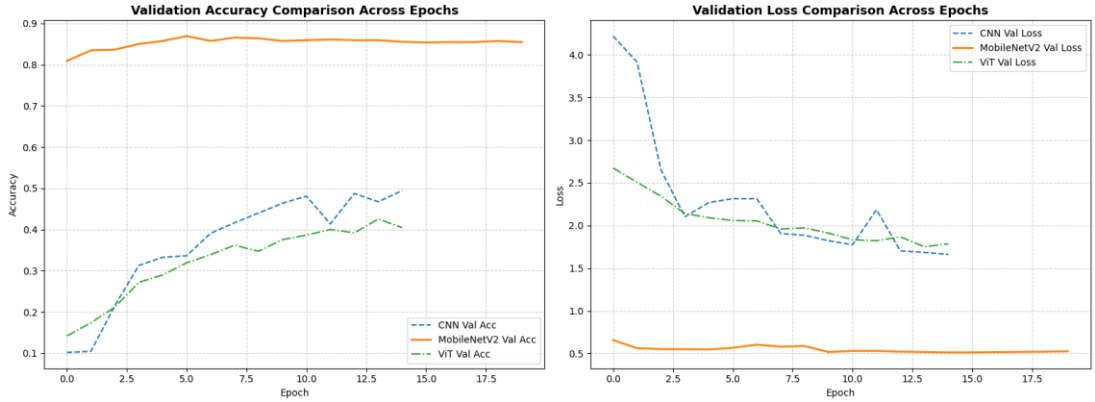


Figure 6.3. Saved neural-model validation accuracy and loss curves.

The curves describe the recorded training runs. They are not repeated-seed uncertainty estimates or deployed inference measurements. Source: saved notebook output [1].

6.3 User or stakeholder validation

No participant study, expert answer-rating exercise, or tourist satisfaction dataset was available. The assistant therefore has no measured answer-correctness, helpfulness, cultural-appropriateness, or hallucination rate in this report. A future evaluation should score factual questions against dated metadata and independently assess unsupported questions, ambiguous identities, and provider failures.

6.4 Integration results

Two deployment consistency gaps are visible in the source. The notebook trains the SVM on features from a fine-tuned MobileNetV2, while the server reconstructs an ImageNet-only backbone. The same dimensionality does not establish the same feature distribution. The saved scaler and classifier should be paired with the exact training extractor before transferring validation claims to serving.

The notebook export includes MobileNetV2 preprocessing expecting the original pixel scale. The backend TFLite helper first divides pixels by 255. If the deployed artefact corresponds to that export, the extra normalisation changes its effective input distribution. This is a code-based compatibility concern, not a measured failure from a live request.

The serving strings state 92.56 percent SVM accuracy and 85.32 percent TFLite accuracy, whereas the notebook comparison gives 93.18 and 85.55 percent for the SVM and fine-tuned neural model. The exported quantized model has no separate recorded evaluation in the inspected notebook. These values must remain distinct.

6.5 Summary of findings

The evidence supports a four-model validation comparison and an implemented recognition-to-assistant workflow. It supports SVM as the strongest recorded classifier configuration in that notebook. It does not establish accuracy of the currently deployed pipeline, independent open-world recognition, or assistant answer quality. The main next step is a reproducible serving evaluation with identical preprocessing and feature extraction.

7 DISCUSSION

7.1 Interpretation of findings

The transfer-learning configurations outperform the custom CNN and native ViT in the recorded experiment. Reusing pretrained visual features is a plausible explanation, but architecture, optimisation, and pretraining change together. An ablation would be needed to isolate their contributions. The SVM result indicates that a separate margin-based classifier can work well on this learned representation; it does not show that SVMs universally outperform neural heads.

7.2 Comparison with literature baselines and existing solutions

The MobileNetV2 design follows a compact convolutional approach [4], and the SVM provides a conventional classifier on extracted features [5]. The ViT literature demonstrates the importance of training scale and transfer [6]. The project's native ViT configuration is materially different from those large-scale settings. The results should therefore be compared within the project experiment and not against unrelated published percentages.

7.3 Research contribution

The research contribution is the implemented comparison of four recognition configurations for the selected landmark classes and the examination of how recognition connects to assistance. The notebook preserves class-level reports, curves, and export steps. The work does not establish a new vision architecture, a new SVM optimisation method, or a new graph retrieval algorithm.

7.4 Product contribution

The product contribution is a connected interface for selecting a photograph, obtaining landmark context, and asking follow-up questions. Structured fields make useful information available even without a generated answer. The local fallback provides a limited response path when Gemini is unavailable. Its utility still depends on recognition correctness, mapping quality, and maintained metadata.

7.5 Limitations and threats to validity

Internal validity is limited by validation reuse, unverified duplicate control, the directory-count discrepancy, and absence of repeated runs. External validity is limited by 17 supported classes, undocumented capture conditions, and no independent photographs or user study. Class confidence is not calibrated open-set reliability.

Deployment validity is limited by extractor and preprocessing differences and inconsistent accuracy labels. Assistant validity is limited by first-match retrieval, incomplete or dated metadata, lack of measured factuality, and no explicit response field identifying whether Gemini or the local fallback answered. Empty strings can also bypass intended fallback defaults, leaving incomplete information.

7.6 Implications

The component can support an initial discovery-and-information workflow, but reliable practical use requires consistent inference artefacts and maintained factual content. The report's most actionable implication is to treat classification, retrieval, and answer generation as separately versioned and tested stages. A correct classifier result should not be used as evidence that every associated visitor fact is current.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The component combines landmark image recognition with structured information and an intelligent assistant. The saved experiment compares four models on 17 classes and records the strongest validation accuracy for deep features with SVM at 93.18 percent. The source implements contextual chat and a local fallback.

The conclusion is bounded by the available evidence. Validation reuse and training-serving inconsistencies prevent the notebook number from being presented as independently verified deployed accuracy. The assistant is implemented, but its answer quality has not been quantified.

8.2 Achievement of objectives

The comparison, inference integration, metadata association, and chat workflow are represented in the available artefacts. Objectives concerning independent recognition performance, unknown-landmark rejection, and user benefit remain only partially evaluated. The next validation phase should use an untouched, provenance-controlled image set and a separate question-answer benchmark.

8.3 Individual research and product contributions

The individual component connects visual classification to landmark information and conversation within the shared tourism platform. Its research outputs are the classifier comparison and stored analysis figures; its product outputs are inference endpoints, metadata mapping, and the recognition and assistant interfaces. Adjacent planning and transport modules remain separate contributions.

8.4 Future directions

First, package the exact fine-tuned feature extractor with the SVM and scaler, verify label order, and align TFLite input preprocessing with its exported graph. Evaluate both serving engines on a frozen independent test set and replace fixed accuracy labels with version-specific evidence.

Second, reconcile dataset counts, document licences, deduplicate images before splitting, and expand geographical and photographic coverage. Tune rejection thresholds on known and unknown images, report calibration and per-class errors, and measure complete end-to-end latency on intended devices.

Finally, add metadata provenance and update dates, disambiguate name matches, expose the response engine, and evaluate assistant factuality with independent reviewers. Graph-based retrieval would be a future extension requiring its own implementation and comparison, rather than a capability inferred from the group title.

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GLOSSARY

Closed-set recognition: Classification among the labels known during training.

Deep feature: A numerical representation produced by a neural network.

Fine-tuning: Updating selected pretrained network weights using task-specific training data.

Quantization: A representation change used to reduce model storage or computation.

Grounding: Supplying retrieved context to guide a generated answer.

Serving parity: Agreement between the transformations and model artefacts used in evaluation and deployment.

APPENDICES

Appendix A Reproducibility notes

The principal experiment is preserved in data-scripts/Landmark_Recognition.ipynb. Its stored outputs provide the tables and figures in Chapter 6. Recognition serving is defined in backend/src/ai_service/landmark_routes.py. The required artefacts include labels.txt, svm_scaler.joblib, svm_landmark_classifier.joblib, and landmark_mobilenet_quantized.tflite.

A reproducible release should additionally preserve the exact feature extractor, dataset manifest, split membership, dependency versions, model hashes, and a dated inference evaluation. The original image collection and an independent test partition were not available for this report.

A future assistant evaluation should include metadata-supported questions, missing facts, ambiguous landmarks, unrelated requests, and provider failures. Record the chosen landmark, retrieved row, response engine, latency, and independent factuality judgement. These are proposed evaluation records, not completed study results.

Appendix B Academic integrity and AI use disclosure

Generative AI assistance through Codex was used to inspect the project notebook and source code, organise this individual report against the supplied thesis template, draft explanations, format references, extract existing notebook figures, and check the rendered document. It did not conduct new training, fabricate application screenshots, or generate participant-study results.

The member should confirm the individual contribution statement before signing. Any additional AI use in dataset preparation, research, or software development should be disclosed in accordance with institutional requirements. The declaration and signatures remain the responsibility of the candidate and supervisors.